Agile Estimation Techniques: Comparing Planning Poker, T-Shirt Sizing, and Story Points

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***Abstract*—Agile teams estimate work so that they can plan a sprint, discuss risk, and set clear expectations. This paper studies three common approaches: Planning Poker, T-shirt sizing, and story points. These approaches are often treated as direct alternatives, but they have different roles. T-shirt sizing is a quick way to group a large backlog. Story points give a closer relative value for stories that may be built soon. Planning Poker is a team process used to discuss different views and agree on an estimate. The paper reviews published studies, compares the three approaches, and proposes a combined method. The method starts with T-shirt sizing, moves selected stories to story points, and uses Planning Poker when estimates differ or important details are unclear. An online shopping example is used to show each step. The results of the comparison show that the methods work best together: T-shirt sizing supports speed, story points support sprint planning, and Planning Poker supports shared understanding. The paper also explains the limits of the comparison and gives a simple plan for testing the method in a real team.**

***Index Terms*—Agile software development, effort estimation, Planning Poker, relative estimation, story points, T-shirt sizing**

I. Introduction

Software teams must plan work before every detail is known. A story that looks small may later need an outside service, more security checks, or extra testing. For this reason, an estimate should show the team's current understanding. It should not be treated as an exact promise.

The Agile Manifesto values people, working software, customer teamwork, and response to change [1]. Its principles also support frequent delivery and regular team review [2]. Estimation fits these ideas when it helps a team ask questions, compare work, and learn from finished stories. It becomes less useful when a number is treated as a fixed deadline or as a measure of one person's output.

Planning Poker, T-shirt sizing, and story points are widely discussed in Agile work. However, they are often explained as if they are the same type of method. This creates a basic problem. T-shirt sizes and story points are scales used to record relative size, while Planning Poker is a team process that can produce an estimate. A team can therefore use Planning Poker to select story points. The three terms are linked, but they are not direct replacements for one another.

This paper addresses that confusion. Its main aim is to explain when each approach is useful and how the three can work together. The paper answers three questions:

1) What does each approach produce?
2) Which planning task does each approach support?
3) How can the approaches be joined in one simple process?

The paper makes three contributions. First, it gives a clear review of earlier studies. Second, it compares all three approaches using the same points: speed, detail, discussion, setup, planning use, and common risk. Third, it proposes a combined flow and applies it to one consistent example. The work is a literature-based comparison, not a claim that one method is always the most accurate.

II. Literature Review

A. Agile Estimation in Practice

Agile estimation is usually relative. Instead of first asking for an exact number of hours, the team compares one story with another. This is useful when the team understands the size difference but cannot yet predict exact time. The values only have meaning inside the team that created them. A five-point story in one team may not match a five-point story in another team.

Usman, Mendes, and Börstler surveyed 60 Agile workers from 16 countries [3]. Planning Poker was reported by 63% of the respondents, while comparison with earlier work was reported by 47%. The study shows that team judgment and relative comparison are common in practice. It also warns that underestimation remains a common problem.

The quality of story information also matters. Pasuksmit, Thongtanunam, and Karunasekera surveyed 121 Agile workers in 25 countries [4]. Most respondents said that written information was important for estimation. Useful items included user stories, acceptance rules, the definition of done, screen designs, and links to other tasks. This finding supports a simple rule: the team should improve an unclear story before trying to give it a more exact value.

B. Studies of Planning Poker

Haugen studied the use of Planning Poker for user-story estimation [5]. Most studied cases improved after the method was introduced, although some cases still had large errors. This means that a clear process can help, but it cannot remove all uncertainty.

Moløkken-Østvold, Haugen, and Benestad compared Planning Poker estimates with separate individual estimates [6]. For the tasks they studied, the group estimates were less optimistic and closer to the final effort than the average of the individual estimates. However, not every part of the comparison gave a clear advantage.

Mahnič and Hovelja studied 13 student teams and compared their estimates with estimates from experienced workers [7]. The student estimates were often too optimistic, and Planning Poker did not remove the effect of limited experience. Their study shows that a meeting process cannot replace technical knowledge.

C. Relative Estimation and Open Problems

Jørgensen and Escott compared relative estimates with time-based estimates in two experiments [8]. Their work treated accuracy, time used, and differences between people as separate questions. The results do not support a simple claim that relative estimates are always faster or more accurate.

Recent reviews also show that estimation problems come from more than the choice of scale. Iqbal *et al.* discuss error and other issues in story-based estimation [9]. Pasuksmit, Thongtanunam, and Karunasekera group common causes of poor estimates into information quality, team factors, estimation practice, project management, and business pressure [10]. Together, these studies show that better stories, team knowledge, and honest discussion are as important as the estimation method.

Table I gives the main lesson taken from each group of studies. The research supports Planning Poker and relative estimation as useful aids, but it does not show one best method for every team. This gap leads to the combined method proposed in this paper.

TABLE I
MAIN LESSONS FROM EARLIER STUDIES

Study area	Main lesson used in this paper
Industry survey [3]	Planning Poker and comparison with past work are common, but low estimates remain a problem.
Story information [4]	Clear stories, acceptance rules, and task links help a team estimate.
Planning Poker [5], [6]	Group discussion can improve estimates, but results still depend on the task.
Team experience [7]	A team process cannot replace skill and past experience.
Relative estimates [8]	Relative values are useful, but they are not always faster or more accurate.
Recent reviews [9], [10]	Poor information, team issues, and business pressure can all reduce quality.

III. STUDY METHOD

This paper uses a structured review and a simple case example. It does not claim to be a full systematic review. The sources were chosen because they directly study Agile values, current estimation practice, story information, Planning Poker, or relative estimation. Journal papers, conference papers, and the official Agile Manifesto were used. General blog posts were not used as research evidence.

The review followed three steps. First, each source was read to find its main result and limit. Second, the three estimation approaches were studied using the same six points shown in Table II. Third, the proposed method was checked with one online shopping backlog. The same stories and values were kept in the text, tables, and figures so that the example remained consistent.

TABLE II
POINTS USED FOR THE COMPARISON

Point	Question asked
Speed	How much team time is normally needed?
Detail	How clearly does the result separate small and large stories?
Discussion	How much team discussion is part of the method?
Setup	What story details, examples, people, or tools are needed?
Planning use	Which backlog or sprint decision does the method support?
Main risk	What common mistake can make the result less useful?

The ratings high, medium, and low are used as clear comparison words. They are not measured scores. For example, a high discussion rating means that discussion is a main part of the method. It does not mean that every meeting will take the same amount of time. This rule avoids giving false accuracy to a literature-based comparison.

The online shopping example is also not presented as an experiment. Its purpose is to test whether the proposed steps work together without conflict. Login is kept as the reference story, payment remains larger because of its security and outside service, and recommendations remains XL because key details are missing. This makes the example easy to follow from the first size to the final decision.

IV. THE THREE ESTIMATION APPROACHES

A. T-Shirt Sizing

T-shirt sizing uses labels such as XS, S, M, L, and XL. It is meant to be broad and fast. A team can use it to group many backlog items before spending time on detailed discussion. The team should agree on the meaning of each size. An XL story should usually be split or studied before it enters a sprint.

The main benefit is speed. The main risk is that labels may mean different things to different people. T-shirt sizes should not be changed directly into hours or points without discussion.

B. Story Points

Story points use relative numbers such as 1, 2, 3, 5, 8, and 13. The number may include work, difficulty, risk, and unclear parts. A known story is used as a reference. For example, if login is 2 points, a search feature with more code and tests may be 5 points.

Points do not mean fixed hours. They should be used within one team. After several sprints, the team can use its own history to help select a fair amount of work. Comparing point totals across teams can give a false result because each team builds its own scale.

C. Planning Poker

Planning Poker is a team process. The story and its acceptance rules are explained first. Each person then chooses a card in private, and all cards are shown together. This reduces the effect of the first answer on other people. When values are far apart, the people with the high and low values explain their reasons. The team clears up the story and votes again.

The value of Planning Poker is not only the final number. Its main value is finding hidden work and different assumptions. The process can take too long if every clear story receives many rounds of voting.

V. Proposed Combined Method

The proposed method uses each approach where it adds the most value. It follows five stages: prepare, classify, refine, resolve, and learn. Table III explains the stages, and Figure 1 shows the full flow.

TABLE III
FIVE STAGES OF THE PROPOSED METHOD

No.	Stage	Action and result
1	Prepare	Check the story goal, acceptance rules, task links, and reference stories.
2	Classify	Give the broad backlog T-shirt sizes; mark XL stories for more work.
3	Refine	Select near-term stories, split large work, and give story points.
4	Resolve	Use Planning Poker when values differ or an important detail is unclear.
5	Learn	Record the result, finish the sprint, and compare the estimate with finished work.

In Stage 1, the team checks whether each story has enough information. The goal is not to finish the design. The goal is to understand the user need, main rules, and known links to other work. A story with missing basic details returns to the Product Owner for clarification.

In Stage 2, the team gives quick T-shirt sizes to the broad backlog. This step helps the team find large and unclear work without holding a long meeting for every item. XL stories are marked for splitting or further study.

In Stage 3, the team selects stories that may be built soon. It chooses one or more reference stories and gives story points to the selected work. In Stage 4, Planning Poker is used only when estimates are far apart or an important concern remains. This rule keeps useful discussion while limiting meeting time.

In Stage 5, the final value and any important assumption are saved. After the sprint, the team checks what changed and whether the reference stories still make sense. This learning step improves the shared scale. It is not used to judge individual workers.

The proposed method differs from using one approach for every item. A large backlog first receives a broad view. Only near-term or unclear stories receive more detail. This design is expected to reduce meeting time while keeping discussion for stories that need it.

VI. Results and Discussion

The results in this section come from the structured comparison and worked example. They are not measurements from a company test. This clear limit prevents the comparison ratings from being presented as proven industry scores.

A. Comparison Results

Table IV compares the three approaches using the same six points. Figure 2 shows how their roles overlap.

The first result is that T-shirt sizing is the best fit for early grouping, where speed matters more than detail. The second result is that story points are more useful for near-term planning because they separate stories more clearly. The third result is that Planning Poker is most useful when the team has different views. It is a process for reaching an estimate, not a separate unit of size.

The main finding is therefore that the three approaches work better as a sequence than as competing choices. T-shirt sizing supports quick team alignment. Story points add a relative number. Planning Poker joins both ideas when the team needs deeper discussion. This relationship is shown in Figure 2.

Table V turns the comparison results into a simple decision guide. It does not state that one method is best in every case. It links each planning need to the approach that most directly supports it.

The guide also shows when not to estimate. If the user goal or acceptance rules are missing, another number will not solve the problem. The correct action is to improve the story or run a short study. In the same way, a team should not use Planning Poker only because it is part of a routine. It should use the method when independent views and discussion can improve understanding.

B. Worked Example

Consider six stories for an online shopping system. The team first gives every story a T-shirt size. It then gives story points only to the stories that are clear enough for near-term work. Table VI shows the result.

Login becomes the two-point reference story. Search, filter, and cart are larger than login but close to one another, so

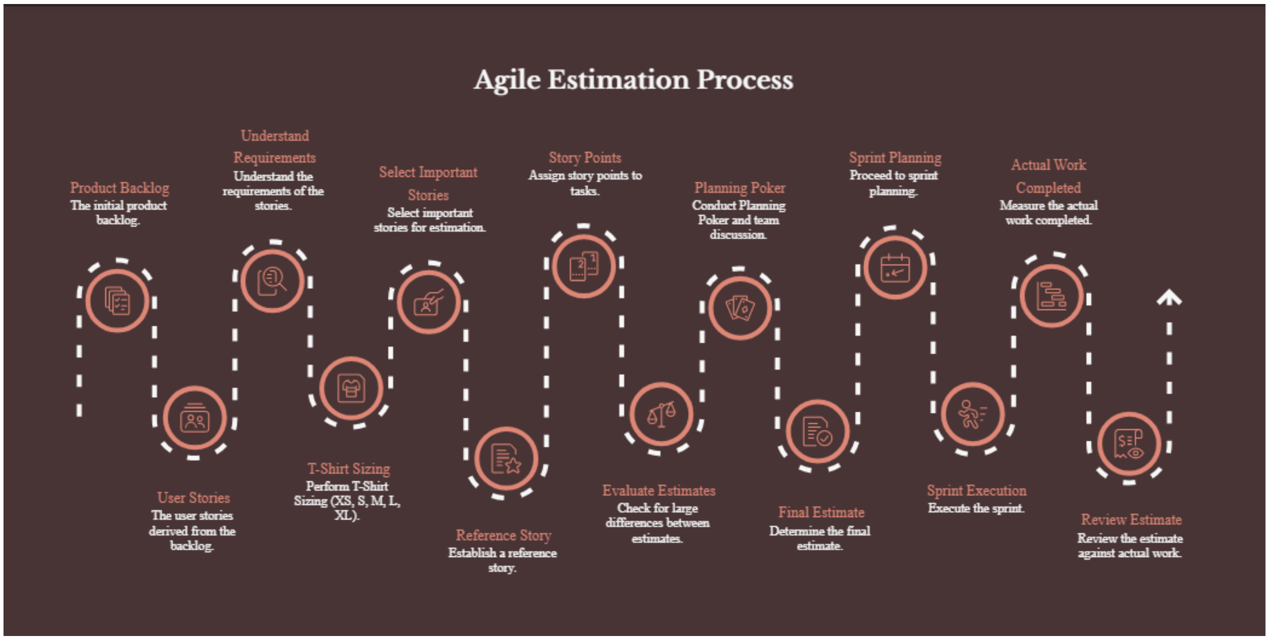

Fig. 1. Proposed Agile estimation process from backlog review to estimate review.

TABLE IV
RESULTS OF THE COMPARISON

Point	**T-shirt sizing**	**Story points**	**Planning Poker**
Type	Broad relative scale	Closer relative number scale	Team estimation process
Output	XS, S, M, L, or XL	A value such as 1, 2, 3, 5, 8, or 13	An agreed value in story points or another scale
Speed	High	Medium	Low to medium
Detail	Low	High	High when discussion is needed
Discussion	Low to medium	Medium	High
Setup	Clear size meanings and example items	Reference stories and a stable point scale	Clear story, team members, cards, and meeting leader
Best use	Early backlog grouping	Near-term review and sprint planning	Finding reasons for different estimates
Main risk	Labels become unclear or are changed directly into numbers	Points are treated as hours or compared across teams	The meeting becomes slow or people agree because of pressure

TABLE V
DECISION GUIDE BASED ON THE RESULTS

Planning need	**Approach to use**
Group a large new backlog quickly	Start with T-shirt sizing.
Prepare clear stories for a sprint	Use story points with reference stories.
Team values are far apart	Use Planning Poker and discuss the reasons.
A story is XL or has missing rules	Split or study the story before pointing it.
Plan with past sprint data	Use the same team's story-point history.
Review an estimate after delivery	Compare the assumptions with finished work.

TABLE VI
EXAMPLE BACKLOG ESTIMATES

Story	**Size**	**Points**	**Main reason**
User login	S	2	Clear reference story
Product search	M	5	Search rules and result states
Product filter	M	5	Several choices and tests
Shopping cart	M	5	Item, quantity, and state changes
Online payment	L	8	Payment service, security, and failures
Recommendations	XL	—	Data and success rules are unclear

Fig. 2. Relationship between Planning Poker, T-shirt sizing, and story points.

the team gives each one 5 points. Payment receives 8 points because it has an outside service, security needs, and more failure cases. Recommendations remains XL and receives no points because the team does not yet know the data source or success rules.

Suppose the first Planning Poker vote for payment is 5, 8, 8, and 13. The person choosing 5 expects the payment library to handle errors. The person choosing 13 expects custom retry and audit work. The discussion finds the exact difference. The team then clears up the acceptance rules and records an agreed value. This example shows why the reasons behind the cards are more useful than simply taking the average.

C. *Meaning of the Results*

More detail does not always mean more accuracy. A precise number can still be weak when the story is unclear. In the proposed method, an XL size or a wide Planning Poker vote is a signal to ask questions or split the work. It is not a reason to force a number.

The findings also show that team context matters. Research reports useful results from Planning Poker, but it also shows the effect of experience and information quality [5], [7], [10]. Teams should keep a few reference stories, record important assumptions, and review finished work. They should not compare points across teams or use velocity as an individual performance target.

VII. Practical Use and Future Testing

For practical use, the team should follow four simple rules. First, state why an estimate is needed. Second, include the people who will build and test the story. Third, stop repeated voting when the real problem is missing information. Fourth, review the estimate after the sprint without blaming individuals.

The proposed method should be tested in a real team before strong claims are made. A future study could compare the current process with the combined method over several sprints. It could measure participant-minutes per story, the number of stories split after discussion, changes made before sprint planning, planned and finished work, and team understanding.

The study should also record changes in team members, sprint length, story type, and story quality. These factors can change the result. A short meeting is not better if the team misses important work, and a stable estimate is not better if people are afraid to disagree.

VIII. Limitations

This paper is based on published studies and one example. It does not report a new company experiment or a statistical test. The speed and detail ratings in Table IV describe common use and may change with team size, skill, tools, and product type.

T-shirt sizing also has less direct research evidence than Planning Poker. The proposed method must therefore be tested in real projects. Finally, changed requirements can change the actual work. A difference between an estimate and final effort does not always mean that the original estimate was poor.

IX. Conclusion

This paper compared Planning Poker, T-shirt sizing, and story points. The main conclusion is that they solve different parts of the same planning problem. T-shirt sizing gives a quick broad view of the backlog. Story points give a closer relative value for near-term stories. Planning Poker gives the team a clear way to discuss different views and agree on an estimate.

The proposed method joins these strengths in five stages: prepare, classify, refine, resolve, and learn. The online shopping example shows how a story can move from a broad size to a point value and, when needed, to a Planning Poker discussion. The method avoids detailed meetings for every backlog item while keeping team discussion for unclear or high-risk work.

No estimation method can remove all uncertainty. Good results still depend on clear story information, team experience, honest discussion, and review of finished work. Estimates should support planning and learning. They should not be treated as fixed promises or as measures of individual performance.

Acknowledgment

The author thanks the School of Computer Science and Engineering at Lovely Professional University for the academic support provided during this study.

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